The Integral is the Area Under the Curve

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Introduction

Anybody who took calculus in high school was most certainly taught the following. To find the area under the curve $y = f(x)$ over some interval $[a, b]$, one uses the formula

$$A = \int_a^b f(x)\,dx$$

by rote, without question. Under the definition of the Riemann (Darboux) integral, where we are using rectangles to approximate the area under a sufficiently well-behaved function, one can see why one can interpret the integral as such.

In this paper, we shall use measure theory to give an alternative proof.

1 Geometric Preliminaries

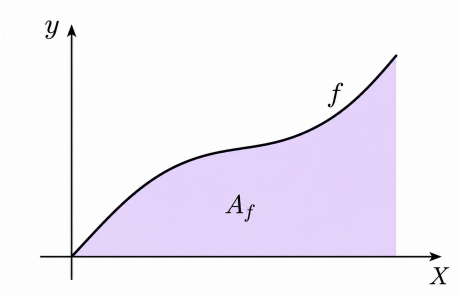


Figure 1: The function f and set A_f. Obviously, the space X is an abstract set; this diagram is only meant to serve as a visual aid. Image generated by ChatGPT 6.

We let $(X, \mathcal{M}, \mu)$ be a σ-finite measure space. It will suffice to prove the desired result for $f \in L^+(\mu)$, that is, for non-negative functions f, for if $f \in L^1(\mu)$, we can simply apply the result to the positive and negative parts of f. To keep things clean, we also assume that $f(x) < \infty$ for all $x \in X$.

The output space of our function is $[0, \infty]$ (we use the extended real number system). Thus, the area bounded by our curve will be a subset of $X \times [0, \infty]$. Denote this set by A_f. Geometrically, for a point $(x, y) \in X \times [0, \infty]$ to be contained in A_f, we must have that $y \leq f(x)$, as illustrated in Figure 1.

Hence, we can define

$$A_f = \{(x, y) \in X \times [0, \infty] : y \leq f(x)\},$$

which is equivalent to writing

$$A_f = \{(x, y) \in X \times [0, \infty] : f(x) - y \geq 0\}.$$

2 Proving the Result

Any hope of proving this result rests on our ability to measure the size of this set.

Lemma 2.1 *The set A_f is $\mathcal{M} \otimes \mathcal{B}_{\mathbb{R}}$-measurable.*

Proof: Define the map $W : X \times [0, \infty] \to [0, \infty]$ by $W(x, y) = f(x) - y$. Then,

$$A_f = W^{-1}([0, \infty]).$$

Thus, if we can show that W is measurable, then we have our result. Indeed, consider the maps $(z, y) \mapsto z - y$ and $(x, y) \mapsto (f(x), y)$. The first map is measurable by continuity, and the second is measurable by Proposition 2.4 in Folland [Fol99]. The composition of measurable functions is measurable, so it follows that W is measurable, and hence that A_f is $\mathcal{M} \times \mathcal{B}_{\mathbb{R}}$-measurable. □

Now that we know A_f is measurable, what is its measure? In the result below, m denotes the standard Lebesgue measure.

Theorem 2.1 (The Integral is the Area Under the Curve.)

$$(\mu \times m)(A_f) = \int f \, d\mu.$$

Proof: σ-finiteness allows us to conclude[1]

$$(\mu \times m)(A_f) = \int m((A_f)_x) \, d\mu(x). \tag{1}$$

Recal that the x-section of A_f is defined as

[1] Theorem 2.36 in [Fol99]

$$(A_f)_x = \{y \in [0, \infty] : (x, y) \in A_f\} = [0, f(x)].$$

Substituting the above equation into Equation 1, and using the properties of the Lebesgue measure, we see that

$$\begin{aligned}(\mu \times m)(A_f) &= \int (f(x) - 0)\, d\mu(x) \\ &= \int f(x)\, d\mu(x),\end{aligned}$$

which is the desired claim. ■

This is for the entire "X-line", so if we want to integrate $f \in L^+(\mu)$ over some set $E \in \mathcal{M}$, we can apply Theorem 2.1 to the function $g := f\chi_E \in L^+(\mu)$, where χ_E denotes the indicator function of the set E, to obtain

$$A_g = \int f\chi_E \, d\mu := \int_E f \, d\mu;$$

The indicator kills the function outside the target interval E. As stated before, for a more general function $f \in L^1(\mu)$, we can apply this result to the positive and negative parts of f, respectively (which are measurable), giving us what we want!

References

[Fol99] Gerald B. Folland. *Real Analysis: Modern Techniques and Their Applications*. 2nd ed. Vol. 234. Pure and Applied Mathematics. New York: John Wiley & Sons, 1999. ISBN: 978-0-471-31716-6.